

13.7 Where the Form Vanishes

Sections 4 and 5 diagonalized a symmetric form and counted the signs on the diagonal. Over $\mathbb{R}$ that is the whole story, but it hides the feature that makes indefinite forms geometrically different from inner products: there are non-zero vectors on which the form vanishes. This section takes those vectors seriously. They are not defects to be quarantined; they organize the space, because each one forces a two-dimensional piece of completely standard shape to split off. The result is a decomposition of any non-degenerate space into a pile of identical planes plus a remainder on which the form vanishes nowhere.

Throughout, F is a field with $\text{char } F \neq 2$ (Definition 0.7.2), V is a finite-dimensional vector space over F with $\dim V = n$, and β is a **symmetric** bilinear form on V (Definition 13.3.1) with quadratic form $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$ (Definition 13.3.2). Two vectors are orthogonal when β pairs them to 0 (Definition 13.4.1). For a subspace $U \subseteq V$ we write, as in Exercise (C2: Orthogonal complements for a form),

$$U^{\perp\beta} = \{ \mathbf{v} \in V : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for every } \mathbf{u} \in U \},$$

and we write $V = U \perp W$ to say that $V = U \oplus W$ with every vector of U orthogonal to every vector of W — an **orthogonal direct sum**. The subscript β is dropped when only one form is in play.

13.7.1 An intuition to give up

Thirteen chapters have trained one reflex above all others: $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ forces $\mathbf{v} = \mathbf{0}$. It is how we proved that an isometry is injective, that an orthogonal list is independent, that a Gram matrix determines its vectors. It is also the first thing that goes.

Take $V = \mathbb{R}^2$ with

$$q(x_1, x_2) = x_1^2 - x_2^2, \quad \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The vector $\mathbf{v} = (1, 1)$ has $q(\mathbf{v}) = 1 - 1 = 0$. It is not $\mathbf{0}$, and the form is not defective: $\det \mathbf{A} = -1 \neq 0$, so β is non-degenerate (Proposition 13.1.2). Indeed $\beta(\mathbf{v}, (1, -1)) = 1 - (1)(-1) = 2 \neq 0$, so $\mathbf{v}$ is perfectly visible to β — it is only invisible to *itself*. The line $\text{span}(\mathbf{v})$ is a line of “length zero” inside a space where length is otherwise perfectly well behaved.

It gets worse, and the worse case is the one worth internalizing. Take $V = \mathbb{R}^4$ with

$$q(\mathbf{x}) = x_1^2 + x_2^2 - x_3^2 - x_4^2,$$

of signature $(2, 2)$ and therefore non-degenerate. Let $W = \text{span}(\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_4)$, a plane. For $\mathbf{w} = a(\mathbf{e}_1 + \mathbf{e}_3) + b(\mathbf{e}_2 + \mathbf{e}_4)$,

$$q(\mathbf{w}) = a^2 + b^2 - a^2 - b^2 = 0,$$

and in fact β vanishes on every pair from W , since $\beta(\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_4) = 0$ too. So β restricted to W is the **zero form** on a two-dimensional subspace of a four-dimensional space on which β is non-degenerate.

Say that again, because it is the sentence an inner-product-trained reader will refuse. A non-degenerate form can vanish identically on a subspace of half the dimension of the whole space. Non-degeneracy is a statement about V ; it says that no vector is orthogonal to *everything*. It

says nothing whatsoever about what β does inside a subspace. The vectors of W above are orthogonal to each other and to themselves; they are not orthogonal to $\mathbf{e}_1 - \mathbf{e}_3$, and that is all non-degeneracy ever promised.

Once that is accepted, the rest of the section is straightforward. The vanishing vectors get a name, and then a structure theorem.

13.7.2 Isotropic vectors and totally isotropic subspaces

An isotropic vector is a non-zero vector that the form cannot tell from zero when it looks at it twice.

Definition 13.7.1 — Isotropic and Anisotropic

Let β be a symmetric bilinear form on V with quadratic form q .

- (a) A vector $\mathbf{v} \in V$ is **isotropic** if $\mathbf{v} \neq \mathbf{0}$ and $q(\mathbf{v}) = 0$, and **anisotropic** if $q(\mathbf{v}) \neq 0$.
- (b) The form β is **anisotropic** if it has no isotropic vectors: that is, if $q(\mathbf{v}) = 0$ implies $\mathbf{v} = \mathbf{0}$. Otherwise β is **isotropic**.

Both clauses hinge on small words. In (a) the vector is required to be **non-zero**, so $\mathbf{0}$ is neither isotropic nor anisotropic; the zero vector always has $q(\mathbf{0}) = 0$ and carries no information. In (b), “anisotropic” is a property of the **form on a particular space**. Anisotropy passes down to subspaces, since a subspace has fewer vectors to test; isotropy does not, and a form isotropic on V is very often anisotropic on a subspace. The word comes from *iso* (equal) and *tropos* (direction): a vector along which the form does not distinguish one direction from another.

Examples, over several fields.

- **A positive definite form has no isotropic vectors.** If $q(\mathbf{v}) > 0$ for $\mathbf{v} \neq \mathbf{0}$ then $q(\mathbf{v}) = 0$ forces $\mathbf{v} = \mathbf{0}$. So every inner product is anisotropic, which is exactly the reflex described above. The same holds for negative definite forms.
- **Over $\mathbb{R}$, signature $(1, 1)$.** For $q = x_1^2 - x_2^2$ the isotropic vectors are the non-zero multiples of $(1, 1)$ and of $(1, -1)$, since $q = (x_1 - x_2)(x_1 + x_2)$. Two lines, and nothing else.
- **Over $\mathbb{C}$.** For $q = x_1^2 + x_2^2$ on $\mathbb{C}^2$ the factorization $q = (x_1 + ix_2)(x_1 - ix_2)$ is available, so $(1, i)$ and $(1, -i)$ are isotropic. Over $\mathbb{R}$ the same q is positive definite. **Isotropy depends on the field**, not only on the polynomial.
- **Over $\mathbb{F}_3$ and $\mathbb{F}_5$.** For $q = x_1^2 + x_2^2$ the squares in $\mathbb{F}_3$ are 0 and 1, so $q(\mathbf{x}) = 0$ needs $x_1 = x_2 = 0$ and the form is anisotropic. In $\mathbb{F}_5$, however, $1^2 + 2^2 = 5 = 0$, so $(1, 2)$ is isotropic. The same formula, two finite fields, two answers.
- **The degenerate case.** Every non-zero vector of the radical is isotropic, since $\mathbf{v} \in \text{rad}(\beta)$ gives $\beta(\mathbf{u}, \mathbf{v}) = 0$ for every $\mathbf{u}$, in particular for $\mathbf{u} = \mathbf{v}$. So a degenerate form is automatically isotropic. The converse fails, and the $\mathbb{R}^2$ example above is the witness.

When a whole subspace consists of isotropic vectors, more is true than the definition says, and it deserves its own name.

Definition 13.7.2 — Totally Isotropic Subspace

A subspace $W \subseteq V$ is **totally isotropic** for β if $\beta(\mathbf{u}, \mathbf{w}) = 0$ for **all** $\mathbf{u}, \mathbf{w} \in W$; equivalently, if $W \subseteq W^{\perp\beta}$. The zero subspace is totally isotropic, trivially.

The two formulations agree because $W \subseteq W^{\perp\beta}$ says exactly that every $\mathbf{w} \in W$ is orthogonal to every $\mathbf{u} \in W$. A third formulation is available in characteristic $\neq 2$ and is the one to use in practice: W is totally isotropic if and only if $q(\mathbf{w}) = 0$ for every $\mathbf{w} \in W$. One direction is immediate; the other is Theorem 13.3.5 applied to the restricted form, which recovers β on W from q on W and so makes it zero. In the $\mathbb{R}^4$ example above we checked exactly this: q vanished on W , and therefore so did β .

Totally isotropic subspaces cannot be too large.

Proposition 13.7.1 — Half the Dimension, at Most

Let β be a non-degenerate symmetric form on V with $\dim V = n$, and let $W \subseteq V$ be totally isotropic. Then $\dim W \leq n/2$.

Proof. Since β is non-degenerate, $\dim W^{\perp\beta} = n - \dim W$ by Exercise (C2: Orthogonal complements for a form) (b). Since W is totally isotropic, $W \subseteq W^{\perp\beta}$, so $\dim W \leq n - \dim W$, which is the claim. ■

Warning. An isotropic vector is not a zero vector, and a form that vanishes on a subspace need not be degenerate. These are two different errors and both are common. The first confuses $q(\mathbf{v}) = 0$ with $\mathbf{v} = \mathbf{0}$; over $\mathbb{R}^2$ with $q = x_1^2 - x_2^2$ the vector $(1, 1)$ separates them. The second confuses “ β vanishes on $W \times W$ ” with “ β vanishes on $\{\mathbf{v}\} \times V$ ”. Degeneracy is about the **second**: the radical consists of vectors killed by pairing with *everything*. The plane W of the $\mathbb{R}^4$ example carries the zero form and contains no radical vector at all.

Quick check. Is the plane $\text{span}(\mathbf{e}_1, \mathbf{e}_2) \subseteq \mathbb{R}^4$ totally isotropic for $q(\mathbf{x}) = x_1^2 + x_2^2 - x_3^2 - x_4^2$? What about $\text{span}(\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 - \mathbf{e}_3)$?

Solution. No to both. On $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$ the form restricts to $x_1^2 + x_2^2$, which is positive definite, so that plane is as far from totally isotropic as possible: it is anisotropic. The second plane is $\text{span}(\mathbf{e}_1, \mathbf{e}_3)$, on which q restricts to $x_1^2 - x_3^2$; its two spanning vectors $\mathbf{e}_1 \pm \mathbf{e}_3$ are indeed isotropic, but $\beta(\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 - \mathbf{e}_3) = 1 - (-1) = 2 \neq 0$, so the plane is not totally isotropic. Containing isotropic vectors is not the same as being totally isotropic.

13.7.3 Splitting off a non-degenerate piece

Everything below rests on one mechanism: if a subspace carries a non-degenerate restriction, it splits off orthogonally and the rest of the space is again non-degenerate. That is the analogue

of “restrict to an invariant complement”, and it is what makes induction possible.

Proposition 13.7.2 — A Non-degenerate Subspace Splits Off

Let β be a non-degenerate symmetric form on V with $\dim V = n$, and let $U \subseteq V$ be a subspace such that $\beta|_{U \times U}$ is non-degenerate. Then

$$V = U \perp U^{\perp\beta},$$

and β restricted to $U^{\perp\beta}$ is non-degenerate.

Idea. Two things to check and one dimension count. The intersection $U \cap U^{\perp\beta}$ is precisely the radical of the restricted form, which the hypothesis kills; the dimension of $U^{\perp\beta}$ is $n - \dim U$ because the global form is non-degenerate; and a vector of $U^{\perp\beta}$ orthogonal to $U^{\perp\beta}$ is orthogonal to all of V , hence zero.

Proof. Write $U^\perp = U^{\perp\beta}$. A vector $\mathbf{v} \in U \cap U^\perp$ lies in U and satisfies $\beta(\mathbf{u}, \mathbf{v}) = 0$ for every $\mathbf{u} \in U$, so $\mathbf{v} \in \text{rad}(\beta|_{U \times U})$, which is $\{\mathbf{0}\}$ by hypothesis (Definition 13.1.4). Hence $U \cap U^\perp = \{\mathbf{0}\}$ and the sum $U + U^\perp$ is direct (Theorem 1.7.4).

Since β is non-degenerate on V , Exercise (C2: Orthogonal complements for a form) (b) gives $\dim U^\perp = n - \dim U$. So $\dim(U \oplus U^\perp) = n$, and a subspace of V of dimension n is V (Theorem 1.6.8). The sum is orthogonal by the definition of $U^\perp$ together with the symmetry of β , so $V = U \perp U^\perp$.

Finally let $\mathbf{w} \in U^\perp$ satisfy $\beta(\mathbf{x}, \mathbf{w}) = 0$ for every $\mathbf{x} \in U^\perp$. For $\mathbf{u} \in U$ we also have $\beta(\mathbf{u}, \mathbf{w}) = 0$, since $\mathbf{w} \in U^\perp$. Every $\mathbf{v} \in V$ is $\mathbf{u} + \mathbf{x}$ with $\mathbf{u} \in U$ and $\mathbf{x} \in U^\perp$, so $\beta(\mathbf{v}, \mathbf{w}) = 0$ for every $\mathbf{v} \in V$, that is $\mathbf{w} \in \text{rad}(\beta) = \{\mathbf{0}\}$. This proves the proposition. ■

13.7.4 Hyperbolic planes

Which two-dimensional forms are isotropic? Over $\mathbb{R}$ with signature $(1, 1)$ we found two isotropic lines, and the example $q = x_1^2 - x_2^2$ can be rewritten by the substitution $u = x_1 + x_2$, $w = (x_1 - x_2)/2$ as $q = 2uw$ — the cross term alone, with no squares at all. That shape is the model, and it is the same over every field.

Definition 13.7.3 — Hyperbolic Pair, Hyperbolic Plane

Let β be a symmetric bilinear form on V . A **hyperbolic pair** is an ordered pair $(\mathbf{e}, \mathbf{f})$ of vectors of V with

$$q(\mathbf{e}) = 0, \quad q(\mathbf{f}) = 0, \quad \beta(\mathbf{e}, \mathbf{f}) = 1.$$

The subspace $H = \text{span}(\mathbf{e}, \mathbf{f})$ is then called a **hyperbolic plane**, and $(\mathbf{e}, \mathbf{f})$ is a **hyperbolic basis** of it.

Four things follow at once. First, $\mathbf{e}$ and $\mathbf{f}$ are linearly independent: if $\mathbf{f} = c\mathbf{e}$ then $\beta(\mathbf{e}, \mathbf{f}) =$

$cq(\mathbf{e}) = 0 \neq 1$. So H really is a plane, $\dim H = 2$. Second, in the basis $(\mathbf{e}, \mathbf{f})$ the Gram matrix of $\beta|_H$ is

$$[\beta|_H]_{(\mathbf{e}, \mathbf{f})} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

whose determinant is $-1 \neq 0$, so $\beta|_H$ is **non-degenerate** (Proposition 13.1.2) and Proposition 13.7.2 applies to it. Third, $q(a\mathbf{e} + b\mathbf{f}) = 2ab$, so on a hyperbolic plane q takes **every** value in F : given $c \in F$, take $a = c/2$ and $b = 1$. Fourth, over $\mathbb{R}$ the Gram matrix has eigenvalues 1 and -1 , so a hyperbolic plane has signature $(1, 1)$ and is the unique real example up to congruence.

Now the workhorse of this section and the next. It says that one isotropic vector is enough to manufacture a whole hyperbolic plane around it.

Lemma 13.7.3 — Every Isotropic Vector Lies in a Hyperbolic Plane

Let $\text{char } F \neq 2$, let β be a **non-degenerate** symmetric bilinear form on V , and let $\mathbf{e} \in V$ be isotropic. Then there is $\mathbf{f} \in V$ such that $(\mathbf{e}, \mathbf{f})$ is a hyperbolic pair. In particular $\mathbf{e}$ lies in a hyperbolic plane $H = \text{span}(\mathbf{e}, \mathbf{f})$, and $\beta|_H$ is non-degenerate.

Idea. Non-degeneracy hands us some $\mathbf{u}$ with $\beta(\mathbf{e}, \mathbf{u}) \neq 0$, and rescaling makes $\beta(\mathbf{e}, \mathbf{u}) = 1$. Only one condition is still missing, $q(\mathbf{u}) = 0$, and $\mathbf{u}$ can be corrected without disturbing what has been achieved: adding a multiple of $\mathbf{e}$ changes $q(\mathbf{u})$ but not $\beta(\mathbf{e}, \mathbf{u})$, because $q(\mathbf{e}) = 0$. Solving for the multiple is one line, and it divides by 2 — which is where the characteristic hypothesis is spent, for the third time in this chapter.

Proof. Since $\mathbf{e} \neq \mathbf{0}$ and β is non-degenerate, there is $\mathbf{u}_0 \in V$ with $c = \beta(\mathbf{e}, \mathbf{u}_0) \neq 0$ (Definition 13.1.4). Put $\mathbf{u} = c^{-1}\mathbf{u}_0$, so that

$$\beta(\mathbf{e}, \mathbf{u}) = 1.$$

Now set

$$\mathbf{f} = \mathbf{u} - \frac{1}{2}q(\mathbf{u})\mathbf{e},$$

which is legitimate because $\text{char } F \neq 2$ makes $2 = 1+1$ invertible. Then, using bilinearity, symmetry and $q(\mathbf{e}) = 0$,

$$\beta(\mathbf{e}, \mathbf{f}) = \beta(\mathbf{e}, \mathbf{u}) - \frac{1}{2}q(\mathbf{u})q(\mathbf{e}) = 1 - 0 = 1,$$

and

$$\begin{aligned} q(\mathbf{f}) &= q(\mathbf{u}) - 2 \cdot \frac{1}{2}q(\mathbf{u})\beta(\mathbf{u}, \mathbf{e}) + \frac{1}{4}q(\mathbf{u})^2q(\mathbf{e}) \\ &= q(\mathbf{u}) - q(\mathbf{u}) + 0 = 0. \end{aligned}$$

Together with $q(\mathbf{e}) = 0$, this shows that $(\mathbf{e}, \mathbf{f})$ is a hyperbolic pair. As noted after Definition 13.7.3, $\mathbf{e}$ and $\mathbf{f}$ are independent, so $H = \text{span}(\mathbf{e}, \mathbf{f})$ is a plane, and the Gram matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is invertible, so $\beta|_H$ is non-degenerate. This proves the lemma. ■

Remark. Non-degeneracy of β is not decoration here. On $\mathbb{R}^2$ with $q(\mathbf{x}) = x_1^2$, the vector $\mathbf{e}_2$ is isotropic and lies in no hyperbolic plane, because $\beta(\mathbf{e}_2, \mathbf{v}) = 0$ for every $\mathbf{v}$ and a hyperbolic pair needs $\beta(\mathbf{e}, \mathbf{f}) = 1$. What fails is the very first line of the proof.

13.7.5 The Witt decomposition

Peeling off a hyperbolic plane and repeating is now a two-line induction, and it is the structural theorem of the section: it does for indefinite forms what “peel off an eigenvector” did for operators in Chapter 8.

Theorem 13.7.4 — Witt Decomposition

Let $\text{char } F \neq 2$ and let β be a non-degenerate symmetric bilinear form on a finite-dimensional V . Then there are an integer $m \geq 0$, hyperbolic planes $H_1, \dots, H_m \subseteq V$ and a subspace $V_0 \subseteq V$ such that

$$V = H_1 \perp H_2 \perp \cdots \perp H_m \perp V_0$$

and β restricted to V_0 is **anisotropic**. In particular $\dim V = 2m + \dim V_0$.

Idea. Induction on $\dim V$. Either β is anisotropic, in which case take $m = 0$ and $V_0 = V$ and stop; or there is an isotropic vector, in which case [Lemma 13.7.3](#) wraps a hyperbolic plane H_1 around it, [Proposition 13.7.2](#) peels H_1 off with a non-degenerate remainder $H_1^\perp$ of dimension $\dim V - 2$, and the induction hypothesis finishes the job there.

Proof. Induct on $\dim V$. If $\dim V = 0$, take $m = 0$ and $V_0 = \{0\}$, on which the form is anisotropic vacuously. Let $\dim V \geq 1$ and suppose the statement holds for every non-degenerate symmetric form on a space of smaller dimension.

If β is anisotropic, take $m = 0$ and $V_0 = V$.

Otherwise β has an isotropic vector $\mathbf{e}$. By [Lemma 13.7.3](#) there is $\mathbf{f}$ with $(\mathbf{e}, \mathbf{f})$ a hyperbolic pair, and $H_1 = \text{span}(\mathbf{e}, \mathbf{f})$ is a hyperbolic plane with $\beta|_{H_1}$ non-degenerate. By [Proposition 13.7.2](#),

$$V = H_1 \perp H_1^{\perp\beta},$$

and β restricted to $V' = H_1^{\perp\beta}$ is non-degenerate, with $\dim V' = \dim V - 2 < \dim V$. The induction hypothesis applied to $(V', \beta|_{V'})$ gives hyperbolic planes $H_2, \dots, H_m \subseteq V'$ and an anisotropic $V_0 \subseteq V'$ with $V' = H_2 \perp \cdots \perp H_m \perp V_0$. Each H_i with $i \geq 2$ and V_0 lie inside $V' = H_1^{\perp\beta}$, so all the summands are pairwise orthogonal, and

$$V = H_1 \perp H_2 \perp \cdots \perp H_m \perp V_0.$$

Counting dimensions, $\dim V = 2m + \dim V_0$. This completes the induction. ■

Two questions are now unavoidable: is m the same for every such decomposition, and is V_0 determined? The first has an answer here, and it comes from a quantity that mentions no decomposition at all.

Definition 13.7.4 — Witt Index

Let β be a non-degenerate symmetric bilinear form on a finite-dimensional V . The **Witt**

index of β is

$$\text{ind}(\beta) := \max\{\dim W : W \subseteq V \text{ totally isotropic}\},$$

the largest dimension of a totally isotropic subspace. By [Proposition 13.7.1](#) it satisfies $\text{ind}(\beta) \leq n/2$, and it is 0 exactly when β is anisotropic.

The maximum exists because the dimensions in question form a non-empty finite set of integers: $W = \{0\}$ is always totally isotropic, and all dimensions are at most n . Nothing in the definition refers to a basis or a decomposition, so the Witt index is an invariant of (V, β) by construction. It is also exactly the number of planes.

Proposition 13.7.5 — The Number of Hyperbolic Planes Is the Witt Index

In any decomposition $V = H_1 \perp \cdots \perp H_m \perp V_0$ as in [Theorem 13.7.4](#), $m = \text{ind}(\beta)$. In particular m is the same for every such decomposition, and so is $\dim V_0 = n - 2m$.

Idea. One inequality is a construction: the first vectors $\mathbf{e}_1, \dots, \mathbf{e}_m$ of the m hyperbolic bases span a totally isotropic subspace of dimension m , since each is isotropic and different planes are orthogonal. The other inequality is an induction that trades one plane for one dimension: given a totally isotropic W , the vectors of W orthogonal to $\mathbf{e}_1$ project into $H_1^\perp$ as a totally isotropic subspace of dimension at least $\dim W - 1$. Repeating m times lands inside the anisotropic V_0 , where the only totally isotropic subspace is $\{0\}$.

Proof. Let $(\mathbf{e}_i, \mathbf{f}_i)$ be a hyperbolic basis of H_i for each i .

Step 1: $\text{ind}(\beta) \geq m$. Put $W = \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_m)$. The $\mathbf{e}_i$ lie in different summands of a direct sum and are non-zero, so they are independent and $\dim W = m$. For $i \neq j$ we have $\beta(\mathbf{e}_i, \mathbf{e}_j) = 0$ because $H_i \perp H_j$, and $\beta(\mathbf{e}_i, \mathbf{e}_i) = q(\mathbf{e}_i) = 0$. By bilinearity β vanishes on $W \times W$, so W is totally isotropic of dimension m .

Step 2: a plane can be traded for a dimension.

Claim. Let β be non-degenerate on V , let $H = \text{span}(\mathbf{e}, \mathbf{f}) \subseteq V$ be a hyperbolic plane, let $V' = H^\perp$, and let $W \subseteq V$ be totally isotropic. Then V' contains a totally isotropic subspace of dimension at least $\dim W - 1$.

Proof. By Proposition 13.7.2, $V = H \perp V'$; let $P: V \rightarrow V'$ be the projection with kernel H , so that $\mathbf{v} - P(\mathbf{v}) \in H$ for every $\mathbf{v}$. Put

$$W_1 = \{ \mathbf{w} \in W : \beta(\mathbf{e}, \mathbf{w}) = 0 \},$$

the kernel of a linear functional on W , so $\dim W_1 \geq \dim W - 1$.

For $\mathbf{w} \in V$ write $\mathbf{w} - P(\mathbf{w}) = a\mathbf{e} + b\mathbf{f}$. Since $\mathbf{e} \in H$ is orthogonal to V' ,

$$\beta(\mathbf{e}, \mathbf{w}) = \beta(\mathbf{e}, a\mathbf{e} + b\mathbf{f}) = a q(\mathbf{e}) + b = b.$$

So $\mathbf{w} \in W_1$ forces $\mathbf{w} - P(\mathbf{w}) \in \text{span}(\mathbf{e})$. Consequently, for $\mathbf{w}, \mathbf{w}' \in W_1$ the vectors $\mathbf{w} - P(\mathbf{w})$ and $\mathbf{w}' - P(\mathbf{w}')$ both lie in $\text{span}(\mathbf{e})$, where β vanishes because $q(\mathbf{e}) = 0$; hence

$$\beta(P(\mathbf{w}), P(\mathbf{w}')) = \beta(\mathbf{w}, \mathbf{w}') - \beta(\mathbf{w} - P(\mathbf{w}), \mathbf{w}' - P(\mathbf{w}')) = 0 - 0 = 0,$$

the first equality using the orthogonality of H and V' . So $P(W_1)$ is a totally isotropic subspace of V' .

It remains to bound its dimension. By Rank–Nullity (Theorem 3.3.1) applied to $P|_{W_1}$, whose kernel is $W_1 \cap H$,

$$\dim P(W_1) = \dim W_1 - \dim(W_1 \cap H).$$

If $W_1 \cap H = \{0\}$, then $\dim P(W_1) = \dim W_1 \geq \dim W - 1$ and we are done. Otherwise pick $\mathbf{x} \neq 0$ in $W_1 \cap H$. Then $P(\mathbf{x}) = 0$, so $\mathbf{x} = \mathbf{x} - P(\mathbf{x}) \in \text{span}(\mathbf{e})$ by the paragraph above, whence $\mathbf{e} \in W$. But then $\beta(\mathbf{e}, \mathbf{w}) = 0$ for every $\mathbf{w} \in W$, because W is totally isotropic and $\mathbf{e} \in W$; so $W_1 = W$. Moreover every $\mathbf{x} \in W_1 \cap H$ lies in $\text{span}(\mathbf{e})$ by the same argument, so $W_1 \cap H = \text{span}(\mathbf{e})$ has dimension 1, and $\dim P(W_1) = \dim W - 1$. Either way the claim holds. ■

Step 3: $\text{ind}(\beta) \leq m$. Let $W \subseteq V$ be totally isotropic. Apply Step 2 to $H = H_1$: since $H_1^{\perp\beta} = H_2 \perp \cdots \perp H_m \perp V_0$ carries a non-degenerate form, we obtain a totally isotropic subspace of it of dimension at least $\dim W - 1$. Repeating with $H_2, \dots, H_m$ in turn produces a totally isotropic subspace of V_0 of dimension at least $\dim W - m$. But $\beta|_{V_0}$ is anisotropic, so a non-zero vector of V_0 has $q \neq 0$ and the only totally isotropic subspace of V_0 is $\{0\}$. Hence $\dim W - m \leq 0$.

Steps 1 and 3 give $\text{ind}(\beta) = m$, and $\dim V_0 = n - 2m$ follows by counting. This proves the proposition. ■

So the number of planes is pinned down. What is not yet pinned down is V_0 itself: different decompositions produce different subspaces, and the claim one wants is that they all carry *the same form* in a suitable sense. Making that precise needs the word “isometry” for forms, and proving it needs Witt’s theorems; both are the business of the next section.

13.7.6 The two familiar fields

Over $\mathbb{R}$ and $\mathbb{C}$ everything can be computed, because the classification is already known.

Proposition 13.7.6 — The Witt Index over the Reals and the Complexes (a)

Let β be a non-degenerate symmetric form on a real V with signature (n_+, n_-) (Definition 13.5.1), so $n_+ + n_- = n$. Then $\text{ind}(\beta) = \min(n_+, n_-)$, and the anisotropic part has dimension $|n_+ - n_-|$ and is definite.

- (b) Let β be a non-degenerate symmetric form on a complex V with $\dim V = n$. Then $\text{ind}(\beta) = \lfloor n/2 \rfloor$. In particular every such form with $n \geq 2$ has an isotropic vector, and the anisotropic part has dimension 0 or 1.

Proof.

- (a) Put $k = \min(n_+, n_-)$. By Theorem 13.4.2 and the rescaling that produced Eq. 13.5.1, together with $n_0 = 0$, there is an orthogonal basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ with $q(\mathbf{v}_i) = 1$ for $i \leq n_+$, $q(\mathbf{v}_i) = -1$ for $n_+ < i \leq n$, and distinct basis vectors orthogonal. For $1 \leq i \leq k$ set

$$\mathbf{e}_i = \mathbf{v}_i + \mathbf{v}_{n_++i}, \quad \mathbf{f}_i = \frac{1}{2}(\mathbf{v}_i - \mathbf{v}_{n_++i}).$$

Then $q(\mathbf{e}_i) = 1 + (-1) = 0$ and $q(\mathbf{f}_i) = \frac{1}{4}(1 + (-1)) = 0$, while

$$\beta(\mathbf{e}_i, \mathbf{f}_i) = \frac{1}{2}(q(\mathbf{v}_i) - q(\mathbf{v}_{n_++i})) = \frac{1}{2}(1 + 1) = 1,$$

so $(\mathbf{e}_i, \mathbf{f}_i)$ is a hyperbolic pair and $H_i = \text{span}(\mathbf{e}_i, \mathbf{f}_i)$ a hyperbolic plane. These planes are pairwise orthogonal, since for $i \neq j$ their spanning vectors involve disjoint sets of the $\mathbf{v}$'s. Let V_0 be the span of the remaining $|n_+ - n_-|$ basis vectors, all of the same sign; then $V = H_1 \perp \dots \perp H_k \perp V_0$, and $\beta|_{V_0}$ is definite, hence anisotropic. By Proposition 13.7.5, $\text{ind}(\beta) = k$.

- (b) By Corollary 13.4.4 and non-degeneracy, there is a basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ with $q(\mathbf{v}_j) = 1$ for every j and distinct basis vectors orthogonal. Write $n = 2k + r$ with $r \in \{0, 1\}$, and let i denote the imaginary unit. For $1 \leq j \leq k$ set

$$\mathbf{e}_j = \mathbf{v}_{2j-1} + i \mathbf{v}_{2j}, \quad \mathbf{f}_j = \frac{1}{2}(\mathbf{v}_{2j-1} - i \mathbf{v}_{2j}).$$

Then $q(\mathbf{e}_j) = 1 + i^2 = 0$ and likewise $q(\mathbf{f}_j) = \frac{1}{4}(1 + i^2) = 0$, while $\beta(\mathbf{e}_j, \mathbf{f}_j) = \frac{1}{2}(1 + 1) = 1$. As in (a) these span k pairwise orthogonal hyperbolic planes, and what is left is the span of $\mathbf{v}_n$ when $r = 1$, on which $q(c\mathbf{v}_n) = c^2 \neq 0$ for $c \neq 0$, and $\{\mathbf{0}\}$ when $r = 0$. So $\text{ind}(\beta) = k = \lfloor n/2 \rfloor$ by Proposition 13.7.5. This proves the proposition. ■

Two readings are worth recording. Over $\mathbb{R}$ the Witt index is the smaller of the two counts of signs: a definite form has index 0, and a form of signature (m, m) has the largest possible index $m = n/2$ — the $\mathbb{R}^4$ example that opened the section, with $n_+ = n_- = 2$, sits exactly there. Over $\mathbb{C}$ the index is as large as it can be for every non-degenerate form, because -1 is a square and so a sum of two squares always factors; the only anisotropic complex spaces are $\{\mathbf{0}\}$ and the lines.

Quick check. A non-degenerate real form has signature $(3, 1)$. What is its Witt index, and what is the anisotropic part?

Solution. The index is $\min(3, 1) = 1$, so exactly one hyperbolic plane splits off and $\dim V_0 = 4 - 2 = 2$. By [Proposition 13.7.6](#) (a), V_0 carries a definite form of dimension $|3 - 1| = 2$, positive definite here since the positive signs are the ones in surplus. Concretely $\mathbb{R}^{3,1}$ with $q = x_1^2 + x_2^2 + x_3^2 - x_4^2$ splits as one hyperbolic plane, spanned by $\mathbf{e}_3 \pm \mathbf{e}_4$ after rescaling, orthogonal to the positive definite plane $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$. The largest totally isotropic subspace is a line, and the lines of this kind sweep out the light cone that Section 11 studies.

13.7.7 Exercises

13.7.7.1 A. Check your understanding

Exercise A1.

- Define **isotropic vector**, **totally isotropic subspace** and **hyperbolic pair**.
- Determine whether the following statement is correct, and justify your answer: a non-degenerate form has no isotropic vectors.
- What is the Witt index of a positive definite form on a space of dimension n ? Of a form of signature $(4, 4)$?
- State the Gram matrix of a hyperbolic plane in a hyperbolic basis, and say why it shows the restricted form is non-degenerate.
- Which hypothesis of [Lemma 13.7.3](#) fails for the form $q(\mathbf{x}) = x_1^2$ on $\mathbb{R}^2$ at the isotropic vector $\mathbf{e}_2$?

Solution.

- A vector $\mathbf{v}$ is isotropic if $\mathbf{v} \neq \mathbf{0}$ and $q(\mathbf{v}) = 0$. A subspace W is totally isotropic if $\beta(\mathbf{u}, \mathbf{w}) = 0$ for all $\mathbf{u}, \mathbf{w} \in W$. A hyperbolic pair is a pair $(\mathbf{e}, \mathbf{f})$ with $q(\mathbf{e}) = q(\mathbf{f}) = 0$ and $\beta(\mathbf{e}, \mathbf{f}) = 1$.
- Incorrect. Non-degeneracy says $\text{rad}(\beta) = \{\mathbf{0}\}$, a statement about pairing with **every** vector. On $\mathbb{R}^2$ with $q = x_1^2 - x_2^2$ the form is non-degenerate, since its Gram matrix has determinant -1 , and $(1, 1)$ is isotropic.
- 0 for a positive definite form, since it is anisotropic, so the only totally isotropic subspace is $\{\mathbf{0}\}$. For signature $(4, 4)$ the index is $\min(4, 4) = 4$ by [Proposition 13.7.6](#) (a), the largest value [Proposition 13.7.1](#) permits in dimension 8.
- It is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, of determinant $-1 \neq 0$, and a form is non-degenerate exactly when its Gram matrix is invertible ([Proposition 13.1.2](#)).
- Non-degeneracy of β . Here $\mathbf{e}_2 \in \text{rad}(\beta)$, so there is no $\mathbf{u}$ with $\beta(\mathbf{e}_2, \mathbf{u}) \neq 0$ and the first line of the proof has nothing to choose.

13.7.7.2 B. Practice

Exercise B1: Find the isotropic vectors.

For each of the following real quadratic forms, find all isotropic vectors, give the signature, and state the Witt index.

- (a) $q(x_1, x_2) = x_1^2 + 2x_1x_2$ on $\mathbb{R}^2$.
- (b) $q(x_1, x_2, x_3) = x_1x_2$ on $\mathbb{R}^3$. (*Is this form non-degenerate?*)
- (c) $q(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2$ on $\mathbb{R}^3$.

Solution.

- (a) $q = x_1(x_1 + 2x_2)$, so $q = 0$ exactly on the two lines $x_1 = 0$ and $x_1 = -2x_2$. The isotropic vectors are the non-zero multiples of $(0, 1)$ and of $(-2, 1)$. The Gram matrix is $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, of determinant $-1 < 0$, so the eigenvalues have opposite signs and the signature is $(1, 1)$. The Witt index is $\min(1, 1) = 1$; indeed $(0, 1)$ is isotropic and by [Lemma 13.7.3](#) the whole space is one hyperbolic plane.
- (b) The Gram matrix is $\begin{pmatrix} 0 & 1/2 & 0 \\ 1/2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, of rank 2, so the form is **degenerate** with $\text{rad}(\beta) = \text{span}(\mathbf{e}_3)$ and the notion of Witt index does not apply as defined. The isotropic vectors are those with $x_1x_2 = 0$, namely the union of the two planes $x_1 = 0$ and $x_2 = 0$, minus $\mathbf{0}$. On the non-degenerate part $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$ the signature is $(1, 1)$ and the index is 1.
- (c) There are none: q is positive definite, so $q(\mathbf{x}) = 0$ forces $\mathbf{x} = \mathbf{0}$. The signature is $(3, 0)$ and the Witt index is 0.

Exercise B2: Build a hyperbolic basis.

- (a) For $q(x_1, x_2) = x_1^2 + x_2^2$ on $\mathbb{C}^2$, find a hyperbolic pair.
- (b) For $q(x_1, x_2, x_3) = x_1^2 + x_2^2 - x_3^2$ on $\mathbb{R}^3$, find a hyperbolic pair $(\mathbf{e}, \mathbf{f})$ with $\mathbf{e} = \mathbf{e}_1 + \mathbf{e}_3$, and identify the anisotropic part $\text{span}(\mathbf{e}, \mathbf{f})^{\perp_\beta}$.

Solution.

- (a) Take $\mathbf{e} = (1, i)$, so $q(\mathbf{e}) = 1 + i^2 = 0$. The vector $\mathbf{u} = (1, -i)$ has $\beta(\mathbf{e}, \mathbf{u}) = 1 \cdot 1 + i \cdot (-i) = 2$, so rescale: $\mathbf{f} = \frac{1}{2}(1, -i)$ gives $\beta(\mathbf{e}, \mathbf{f}) = 1$ and $q(\mathbf{f}) = \frac{1}{4}(1 + i^2) = 0$. So $((1, i), \frac{1}{2}(1, -i))$ is a hyperbolic pair, and $\mathbb{C}^2$ is a single hyperbolic plane.
- (b) $q(\mathbf{e}) = 1 - 1 = 0$, so $\mathbf{e}$ is isotropic. Follow the proof of [Lemma 13.7.3](#): $\mathbf{u}_0 = \mathbf{e}_1$ gives $\beta(\mathbf{e}, \mathbf{e}_1) = 1$, so take $\mathbf{u} = \mathbf{e}_1$, and $q(\mathbf{u}) = 1$, whence

$$\mathbf{f} = \mathbf{e}_1 - \frac{1}{2}(\mathbf{e}_1 + \mathbf{e}_3) = \frac{1}{2}(\mathbf{e}_1 - \mathbf{e}_3).$$

Check: $q(\mathbf{f}) = \frac{1}{4}(1 - 1) = 0$ and $\beta(\mathbf{e}, \mathbf{f}) = \frac{1}{2}(1 + 1) = 1$. For the complement, $\mathbf{v} = (v_1, v_2, v_3)$ is orthogonal to both $\mathbf{e}$ and $\mathbf{f}$ exactly when $v_1 - v_3 = 0$ and $v_1 + v_3 = 0$, that is $v_1 = v_3 = 0$. So $\text{span}(\mathbf{e}, \mathbf{f})^{\perp_\beta} = \text{span}(\mathbf{e}_2)$, on which $q = x_2^2$ is positive definite. This is the decomposition of [Theorem 13.7.4](#) with $m = 1$, matching $\min(2, 1) = 1$.

Exercise B3: A totally isotropic subspace of the largest size.

Let $q(\mathbf{x}) = x_1x_4 + x_2x_3$ on $\mathbb{R}^4$.

- (a) Write down the Gram matrix and verify that β is non-degenerate.
- (b) Exhibit a totally isotropic subspace of dimension 2, and explain why none of dimension 3 exists.
- (c) Determine the signature of q and confirm the Witt index by [Proposition 13.7.6](#) (a).

Solution.

- (a) Halving the cross-term coefficients (Eq. 13.3.1),

$$\mathbf{A} = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

whose determinant is $\frac{1}{16}$ det of the reversal permutation matrix, namely $\frac{1}{16} \cdot 1 \neq 0$. So β is non-degenerate by Proposition 13.1.2.

- (b) Take $W = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$. Every $\mathbf{w} = (a, b, 0, 0)$ has $q(\mathbf{w}) = 0$, and $\beta(\mathbf{e}_1, \mathbf{e}_2) = 0$ from the matrix, so W is totally isotropic of dimension 2. No totally isotropic subspace has dimension 3, because [Proposition 13.7.1](#) caps it at $4/2 = 2$.
- (c) The two planes $\text{span}(\mathbf{e}_1, \mathbf{e}_4)$ and $\text{span}(\mathbf{e}_2, \mathbf{e}_3)$ are orthogonal, and $(\mathbf{e}_1, 2\mathbf{e}_4)$ is a hyperbolic pair, since $q(\mathbf{e}_1) = q(2\mathbf{e}_4) = 0$ and $\beta(\mathbf{e}_1, 2\mathbf{e}_4) = 2 \cdot \frac{1}{2} = 1$; the same works for $(\mathbf{e}_2, 2\mathbf{e}_3)$. So each plane is hyperbolic, of signature $(1, 1)$. Hence q has signature $(2, 2)$ and, by [Proposition 13.7.6](#) (a), Witt index $\min(2, 2) = 2$, attained by the W of part (b).

13.7.7.3 C. Going deeper**Exercise C1: Isotropy makes the form surjective.**

Let β be a non-degenerate symmetric form on V over a field F with $\text{char } F \neq 2$.

- (a) Prove that if β has an isotropic vector, then $q: V \rightarrow F$ is surjective.
- (b) Deduce that if some $c \in F$ is not of the form $q(\mathbf{v})$, then β is anisotropic.
- (c) Use (b) to show that the form $q(x_1, x_2) = x_1^2 + x_2^2$ on $\mathbb{F}_3^2$ is anisotropic, and give its Witt index.

Solution.

- (a) Let $\mathbf{e}$ be isotropic. By [Lemma 13.7.3](#) there is $\mathbf{f}$ with $(\mathbf{e}, \mathbf{f})$ a hyperbolic pair. For $c \in F$, put $\mathbf{v} = \frac{c}{2}\mathbf{e} + \mathbf{f}$; this is legitimate since $\text{char } F \neq 2$. Then

$$q(\mathbf{v}) = \frac{c^2}{4}q(\mathbf{e}) + 2 \cdot \frac{c}{2}\beta(\mathbf{e}, \mathbf{f}) + q(\mathbf{f}) = 0 + c + 0 = c.$$

So every $c \in F$ is a value of q .

- (b) Contrapositive of (a): if q is not surjective then β has no isotropic vector.

- (c) The squares in $\mathbb{F}_3$ are $0^2 = 0$, $1^2 = 1$, $2^2 = 1$, so the values of $x_1^2 + x_2^2$ are 0, 1, 2 — which is all of $\mathbb{F}_3$, so (b) gives nothing directly. Argue instead from the definition: $q(\mathbf{x}) = 0$ needs $x_1^2 = -x_2^2 = 2x_2^2$. If $x_2 \neq 0$ then $x_2^2 = 1$, so $x_1^2 = 2$, which is not a square in $\mathbb{F}_3$; hence $x_2 = 0$ and then $x_1 = 0$. The form is anisotropic, its Witt index is 0, and the Gram matrix $\mathbf{I}_2$ shows it is non-degenerate. Over $\mathbb{F}_5$ the same form is isotropic, since $1^2 + 2^2 = 0$.

Exercise C2: A maximal totally isotropic subspace and its complement.

Let β be non-degenerate symmetric on V with $\dim V = n$, and let $W \subseteq V$ be totally isotropic with $\dim W = \text{ind}(\beta) = m$.

- Prove that $\dim W^{\perp\beta} = n - m$ and $W \subseteq W^{\perp\beta}$.
- Prove that the form induced by β on the quotient $W^{\perp\beta}/W$ is well defined and anisotropic. *Hint: a coset with an isotropic representative would enlarge W .*
- Deduce that $n - 2m = \dim(W^{\perp\beta}/W)$ agrees with the dimension of the anisotropic part in [Theorem 13.7.4](#).

Solution.

- The dimension is Exercise (C2: Orthogonal complements for a form) (b), using non-degeneracy, and the inclusion is [Definition 13.7.2](#).
- For $\mathbf{u}, \mathbf{v} \in W^{\perp\beta}$ and $\mathbf{w}, \mathbf{w}' \in W$,

$$\beta(\mathbf{u} + \mathbf{w}, \mathbf{v} + \mathbf{w}') = \beta(\mathbf{u}, \mathbf{v}),$$

because each of the three remaining terms pairs a vector of W with a vector of $W^{\perp\beta} \supseteq W$, hence vanishes. So $\tilde{\beta}(\mathbf{u} + W, \mathbf{v} + W) := \beta(\mathbf{u}, \mathbf{v})$ is well defined, and it is bilinear and symmetric. Suppose $\tilde{q}(\mathbf{u} + W) = 0$ with $\mathbf{u} \notin W$, that is $q(\mathbf{u}) = 0$ with $\mathbf{u} \in W^{\perp\beta} \setminus W$. Then $W + \text{span}(\mathbf{u})$ is totally isotropic: it is spanned by isotropic vectors, $\beta(\mathbf{u}, \mathbf{w}) = 0$ for $\mathbf{w} \in W$ because $\mathbf{u} \in W^{\perp\beta}$, and β vanishes on $W \times W$. Its dimension is $m + 1 > \text{ind}(\beta)$, a contradiction. So $\tilde{\beta}$ is anisotropic.

- By (a), $\dim(W^{\perp\beta}/W) = (n - m) - m = n - 2m$. By [Proposition 13.7.5](#) the anisotropic part V_0 of a Witt decomposition also has dimension $n - 2m$, since the index is m . (That the two anisotropic forms agree, and not merely their dimensions, is proved in the next section.)

Exercise C3: Where characteristic 2 breaks.

Work over $\mathbb{F}_2$ and let β be the dot product on $\mathbb{F}_2^2$, with Gram matrix $\mathbf{I}_2$.

- Verify that β is symmetric and non-degenerate, and find all $\mathbf{v}$ with $q(\mathbf{v}) = 0$.
- Show that β has a vector with $q(\mathbf{v}) = 0$ and $\mathbf{v} \neq \mathbf{0}$ that lies in no hyperbolic plane, and identify which line of the proof of [Lemma 13.7.3](#) fails.
- Explain in one sentence why the third description of “totally isotropic” given after [Definition 13.7.2](#) is unavailable here.

Solution.

- (a) The Gram matrix $\mathbf{I}_2$ is symmetric and invertible, so β is symmetric and non-degenerate (Proposition 13.1.2). Here $q(\mathbf{x}) = x_1^2 + x_2^2 = (x_1 + x_2)^2$ over $\mathbb{F}_2$, which vanishes exactly when $x_1 = x_2$. So $q(\mathbf{v}) = 0$ for $\mathbf{v} = \mathbf{0}$ and for $\mathbf{v} = (1, 1)$.
- (b) Take $\mathbf{v} = (1, 1)$. A hyperbolic plane containing it would need $\mathbf{f}$ with $q(\mathbf{f}) = 0$ and $\beta(\mathbf{v}, \mathbf{f}) = 1$. The only non-zero vector with $q = 0$ is $\mathbf{v}$ itself, and $\beta(\mathbf{v}, \mathbf{v}) = q(\mathbf{v}) = 0 \neq 1$. The step that fails is the correction $\mathbf{f} = \mathbf{u} - \frac{1}{2}q(\mathbf{u})\mathbf{e}$: the element $2 = 1 + 1$ is 0 in $\mathbb{F}_2$ and is not invertible, so $\frac{1}{2}$ does not exist.
- (c) Because it is Theorem 13.3.5, which needs $\text{char } F \neq 2$ to divide by 2; over $\mathbb{F}_2$ the line $\text{span}((1, 1))$ has $q \equiv 0$ on it and is indeed totally isotropic here, but the implication “ q vanishes on W implies β vanishes on $W \times W$ ” is not available in general.